



— REALITY CAPTURE AS A SERVICE

Walk it before you *book it.*

Every feature of the RCAAS platform in one place: the 3D tour, the studio, the client website, restaurant table booking, branding, privacy and cloud storage.

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languages: English, नेपाली, 中文

7

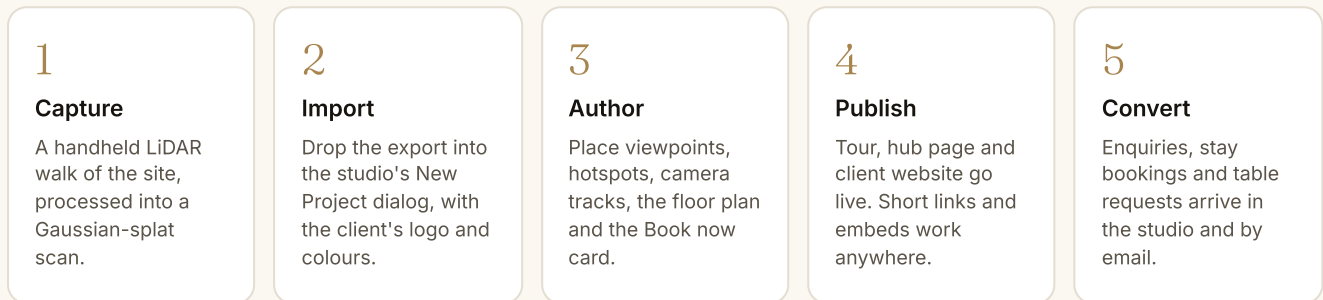
hotspot types

4

ways to move through a scan

One scan. *Years* of visitors.

RCAAS turns a real place (a hotel, a restaurant, a college, a heritage site) into a live 3D tour that opens on any phone. Visitors walk it, then enquire, book a stay or reserve a table without leaving the room.



WHO IT'S FOR

- ◆ **Hotels and restaurants:** rooms, bars and dining rooms guests can walk, then book
- ◆ **Colleges and campuses:** labs, canteens and classrooms for new students
- ◆ **Heritage sites:** stupas and courtyards documented and shared
- ◆ **Offices, showrooms, facilities:** a space a buyer can inspect remotely

THE THREE SURFACES

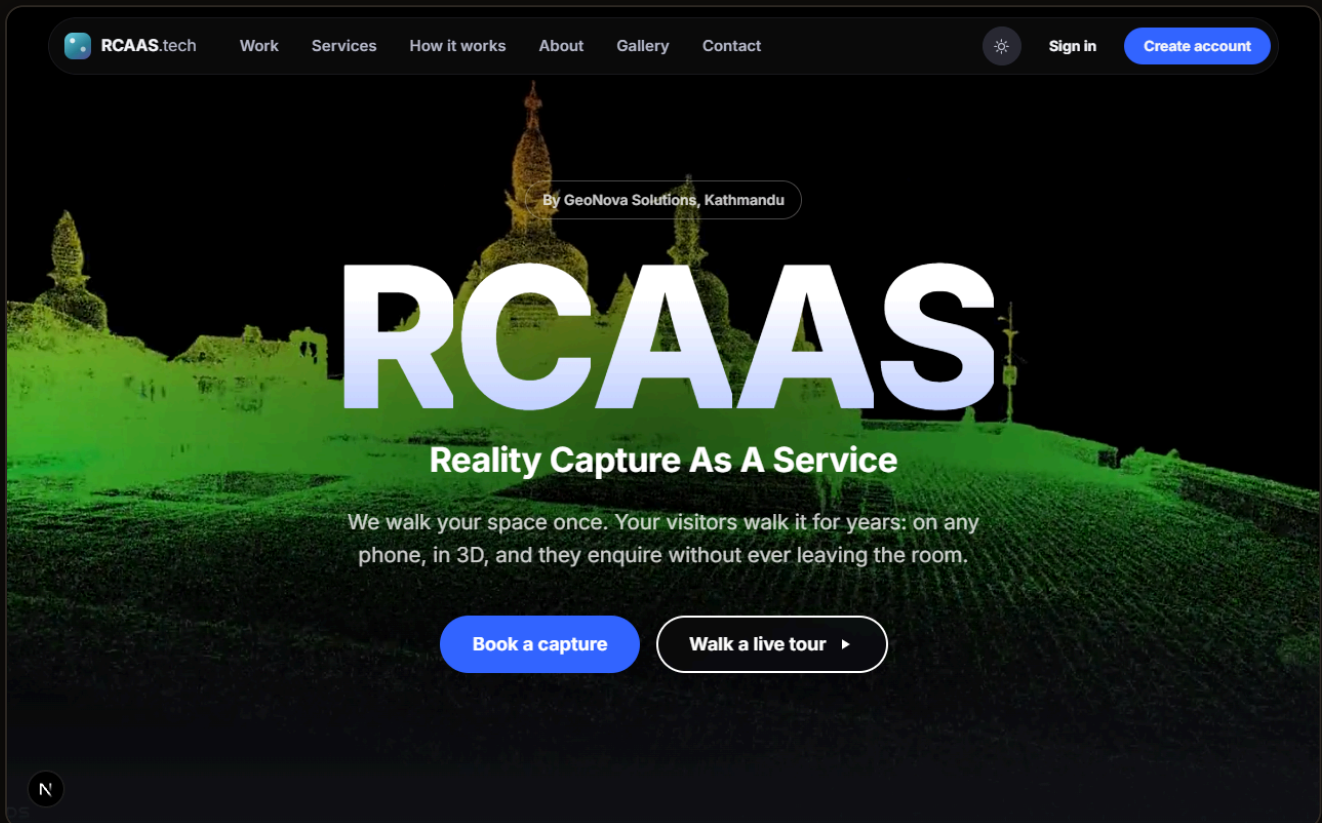
- ◆ **The tour** `/t/<project>` : the 3D experience visitors see
- ◆ **The client website** `/s/<project>` : story, rooms, menu, booking
- ◆ **The studio** `/studio` : where the team builds and runs all of it

We don't sell a 3D model.
We sell *enquiries*.

Every tour carries an enquiry form inside the room, so a visitor who is already looking can ask without leaving.

A first impression that *moves*.

RCAAS.tech is the company's shop window: work, services, how it works, about, gallery and contact, with animation built on GSAP and Framer Motion.

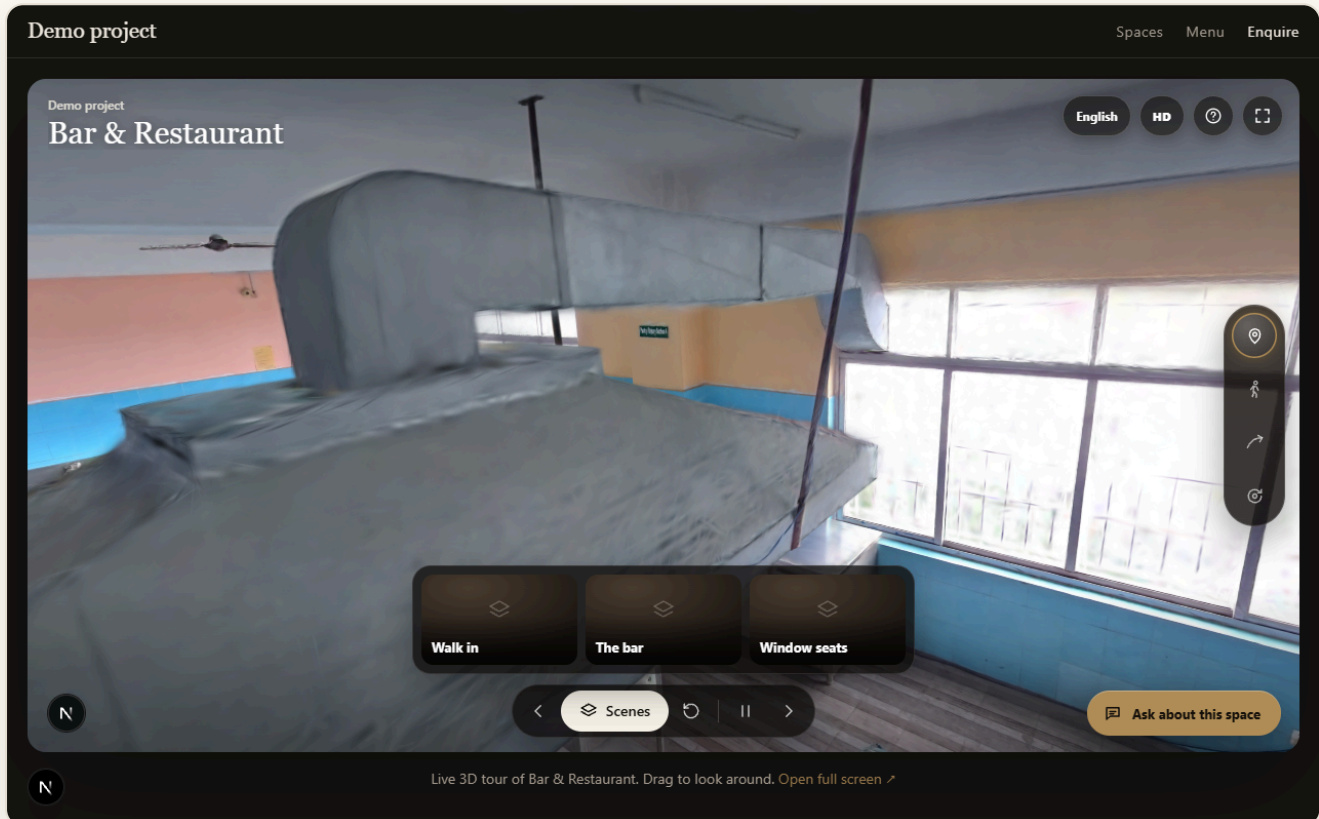


Home. The hero over a point-cloud scan of a Kathmandu stupa.

- ◇ **GSAP motion:** a split-text hero, word-by-word reveals, count-up numbers and parallax
- ◇ **View transitions** between pages, plus a light/dark theme toggle
- ◇ **Reduced motion respected:** visitors who ask for less motion get simple fades
- ◇ **Gallery** of every published tour, live from the API
- ◇ **Accounts:** sign in, create account, profile and settings drawer
- ◇ **Contact:** what to tell us about a space, and every way to reach the team

Walk the room, *exactly* as it is.

Scans render live in the browser with Gaussian splatting. No app, no plugin, and it runs on phones as well as desktops.



The tour. Movement rail on the right, viewpoints at the bottom, "Ask about this space" always in reach.

MOVING AROUND

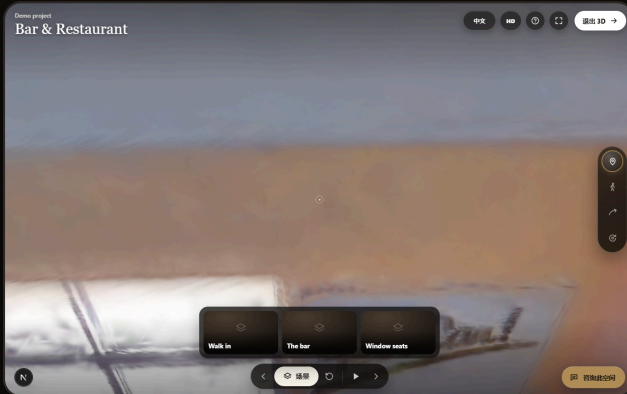
- ◇ **Four modes:** Viewpoints, Walk (first person), Fly and Orbit
- ◇ **Camera tracks:** guided flights with play, pause, next and previous
- ◇ **Floor map** with a live "you are here" marker
- ◇ **Portals** step from one space into the next

HOTSPOTS, SEVEN KINDS

- Text
- Image
- Video
- Audio
- Link
- Portal
- Table

Each hotspot can show all the time or only when the visitor comes near, so the room isn't cluttered.

Day or night. *Any* language.



中文. The whole tour switches language, controls included.



Night. The same room scanned after dark, keeping the visitor's view.

THREE LANGUAGES

English, नेपाली and 中文 from a picker, or set in the link with `?lang=`. A test checks that every label has all three.

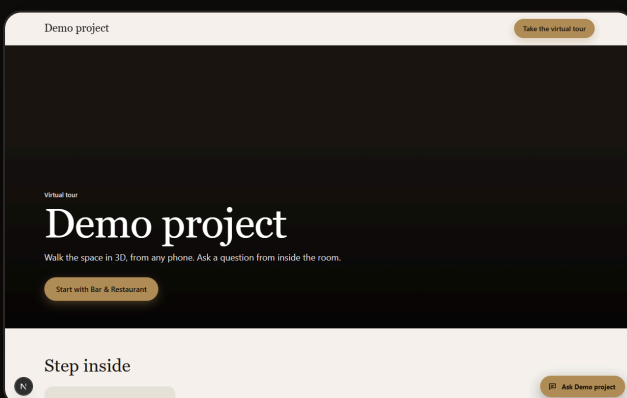
DAY / NIGHT

A space can have a night version. One tap switches between them and keeps the camera where it is.

RIGHT QUALITY

The tour measures the device and picks a quality tier on its own. An HD toggle lets strong devices go further.

Sharing a tour

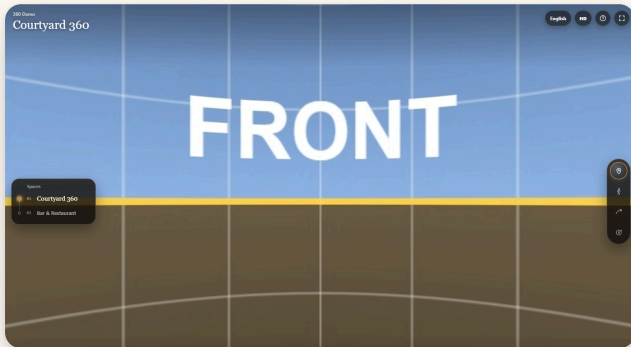


Hub page in the client's brand, with every space one tap away.

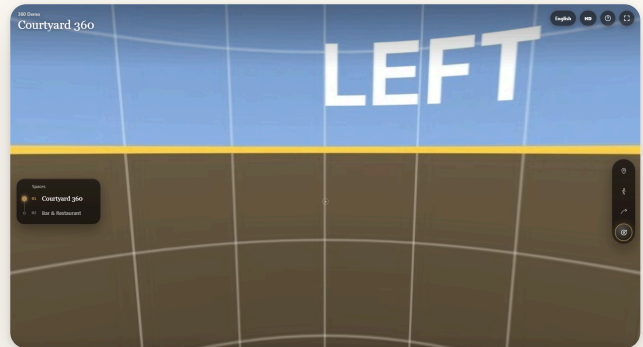
- ◇ **Short links:** `/t/<project>/<space>` for any space
- ◇ **Hub page:** `/t/<project>` lists all of a client's spaces
- ◇ **Embeds** with keys, allowed only on the sites the client lists
- ◇ **Visit stats:** visits and time in the tour, recorded per space

Film it once. Look *everywhere*.

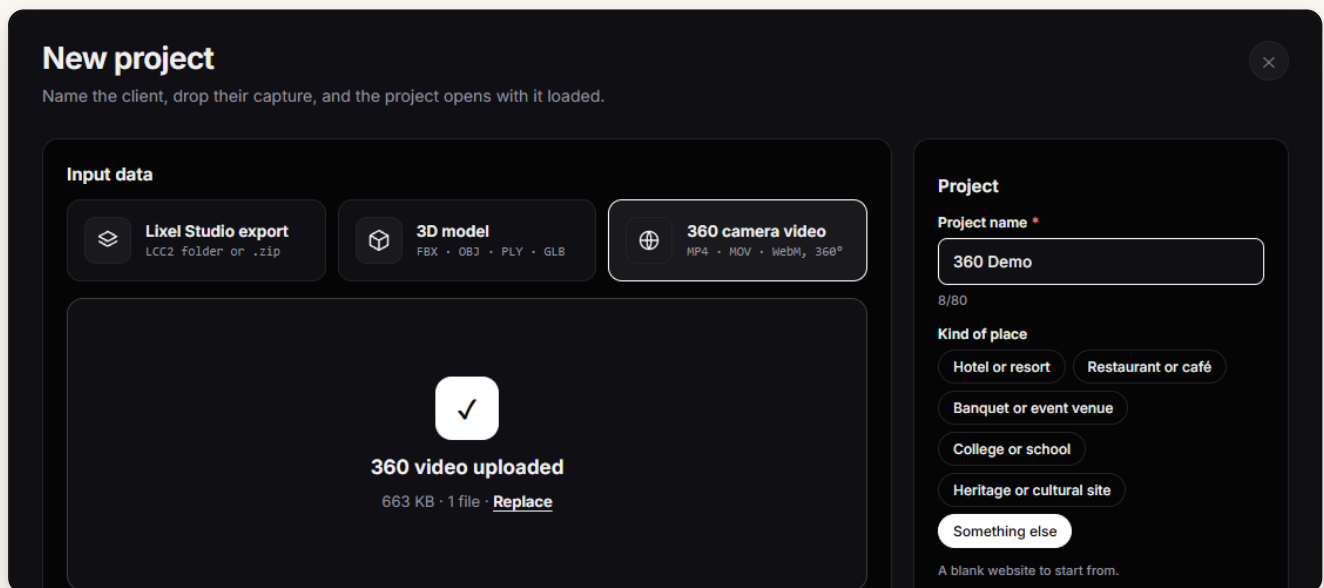
No scanner on site? A 360° camera video becomes a space of its own: visitors stand where the camera stood and look all the way round.



Courtyard 360. The space opens facing the front of the camera, like any scan.



Turned round. Drag in any direction, in every mode; test footage shown.



New project, from a 360 video. Drop one MP4, MOV or WebM; the project opens with the video as its first space.

IN THE STUDIO

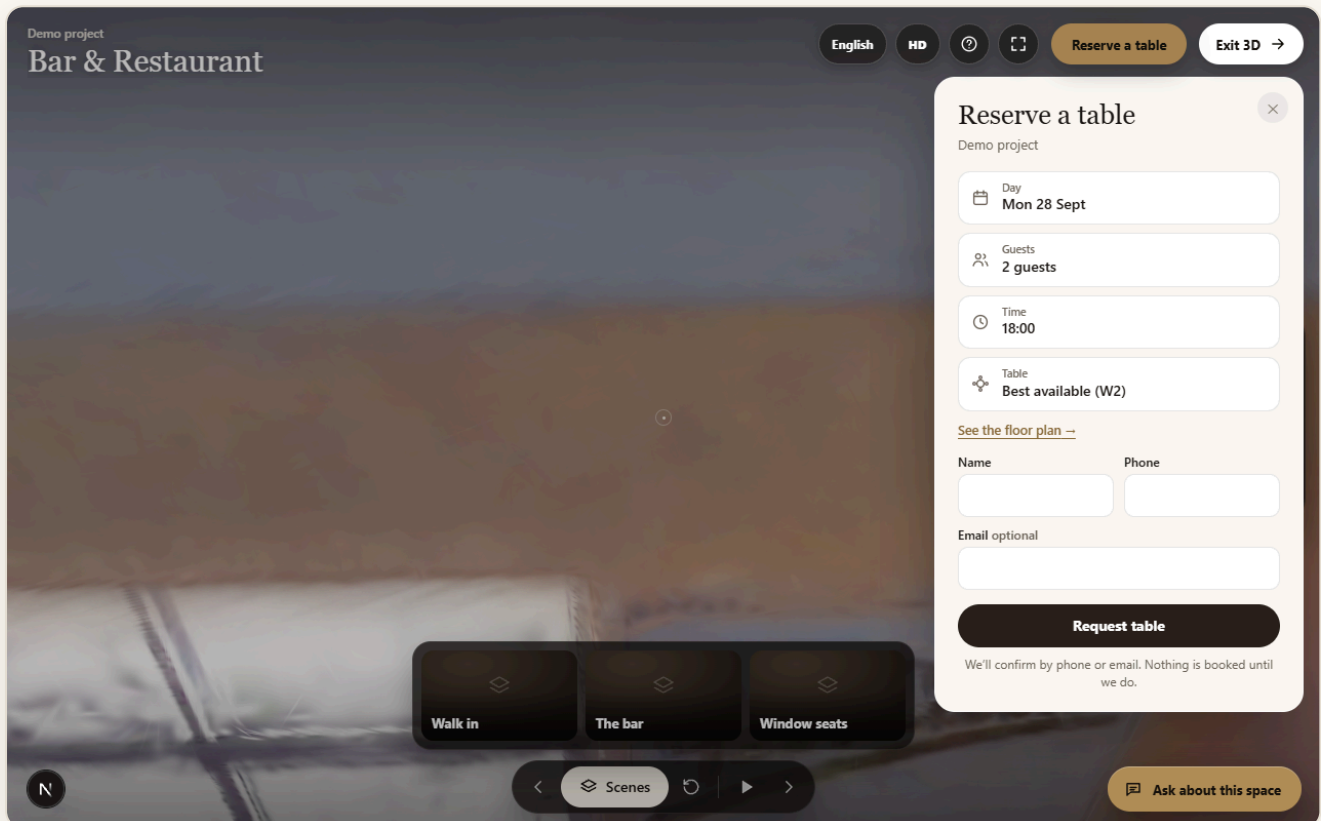
- ◆ **New project** → 360 camera video, or add one to any project from the upload panel
- ◆ **Checked on upload:** the frame must be a 2:1 equirectangular video
- ◆ **Raw INSV files** are refused, with how to export them from Insta360 Studio

IN THE TOUR

- ◆ **Streams** by byte range, so it starts before the whole file arrives
- ◆ **Muted and looping;** pauses when the tab is hidden
- ◆ **Works like any space:** hotspots, the space list, turning it to face the right way

From "wow" to *booked*.

The moment a visitor likes what they see, the next step is one tap away, without leaving the 3D view.



Reserve a table, inside the tour: day, guests, time and the best free table, with the floor plan one link away.

BOOK NOW

A booking card for rooms or stays, set up per space in the studio, linking to the client's own booking page.

ENQUIRE

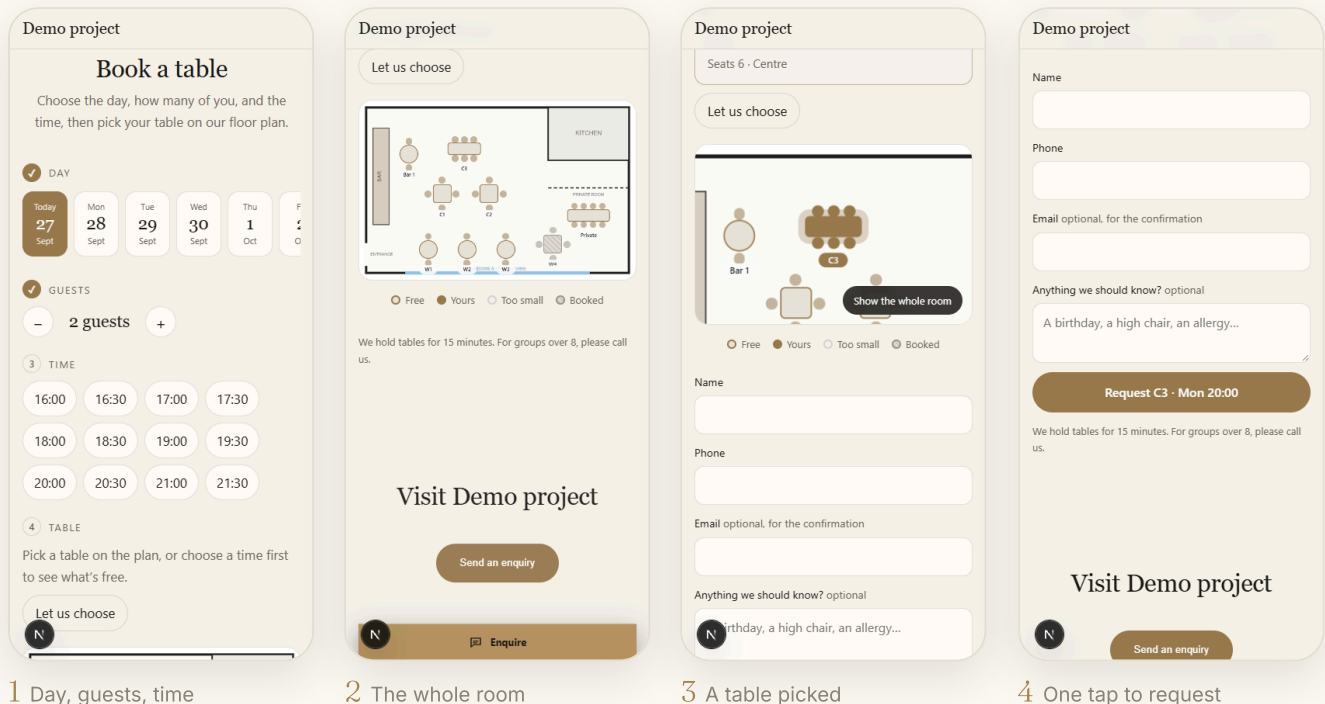
"Ask about this space" sends the space and the last hotspot the visitor looked at, so the team knows what they were asking about.

RESERVE A TABLE

The same booking as the website, as a card in the tour, with a floor-plan overlay and table hotspots placed in the scan.

Pick your table on the *real* floor plan.

Guests don't just pick a time. They see the restaurant's own floor plan and choose the table they want, with GSAP animation guiding each step.



1 Day, guests, time

2 The whole room

3 A table picked

4 One tap to request

HOW A GUEST BOOKS

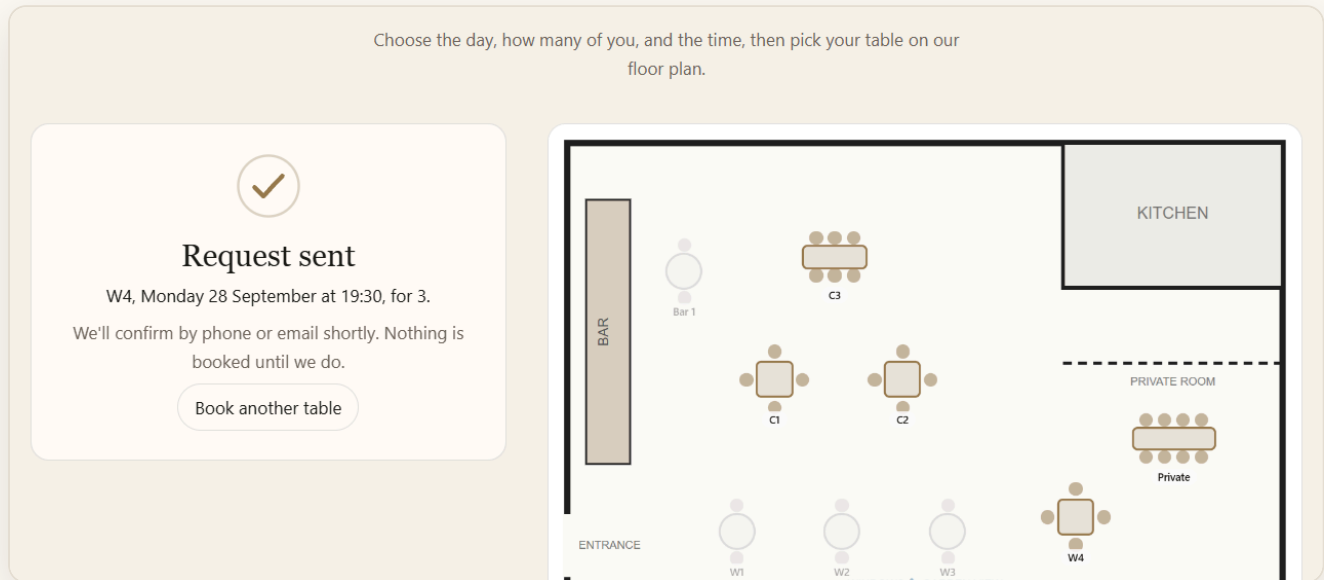
- ◇ **Day, guests, time:** only times with a free table that fits are offered
- ◇ **The plan:** tables show as free, yours, too small or booked; tapping one selects it with an animated highlight
- ◇ **"Let us choose":** the smallest free table that fits the party
- ◇ **"See the view from this table in 3D":** jumps into the live tour at that seat, for tables with a saved view
- ◇ **Details:** name, phone, optional email and a note (birthday, high chair, allergy)
- ◇ **Request sent:** nothing is booked until the restaurant confirms

WHAT THE RESTAURANT SETS UP

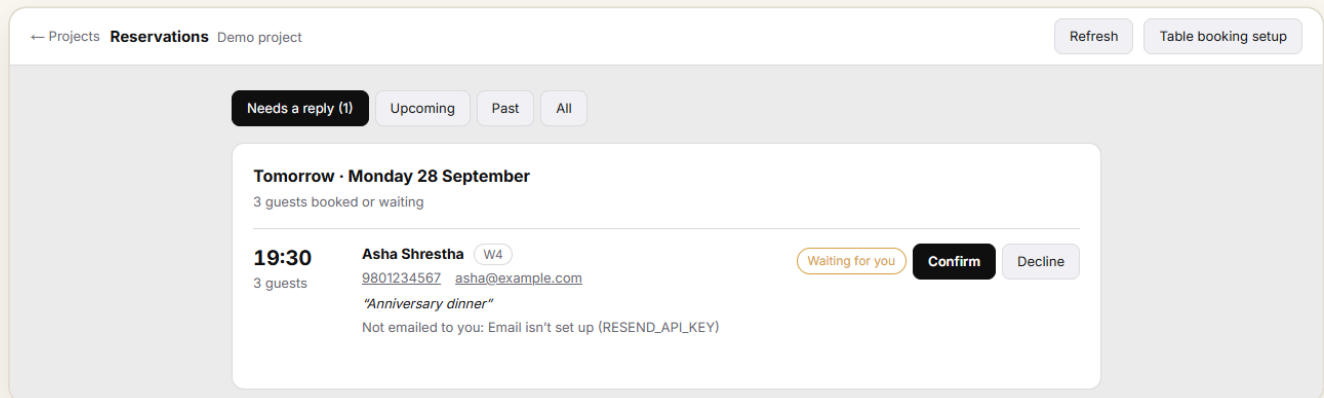
- ◇ **Its floor plan:** upload the plan the hotel provides, then click to place tables and drag to move them
- ◇ **Each table:** name, seats, shape (round, square, long), area and the view from it in the tour
- ◇ **Opening hours, closed days and time zone,** and how long a table is held per booking
- ◇ **Its own note** for guests, such as "groups over 8, please call"

The reservations *inbox*.

Every request lands in the studio, grouped by day, with one-tap Confirm and Decline.



Request sent. The guest sees their table on the plan; the restaurant gets the request.



Reservations in the studio: Needs a reply, Upcoming, Past and All, with guest counts per day.

FOR THE TEAM

Confirm, decline or cancel any request. Phone and email are links. An email alert arrives for each new request (through Resend).

FOR THE GUEST

A **confirmation email** when the table is confirmed, or a note if it's declined. The same booking on the website, the tour card and table hotspots.

BUILT FOR CORRECTNESS

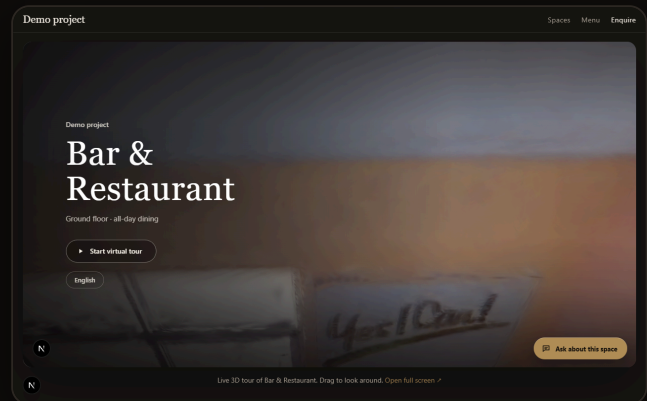
Times follow the restaurant's time zone. A booking holds its table for the whole stay, and requests are handled one at a time, so no table is booked twice.

A whole website, *built around* the scan.

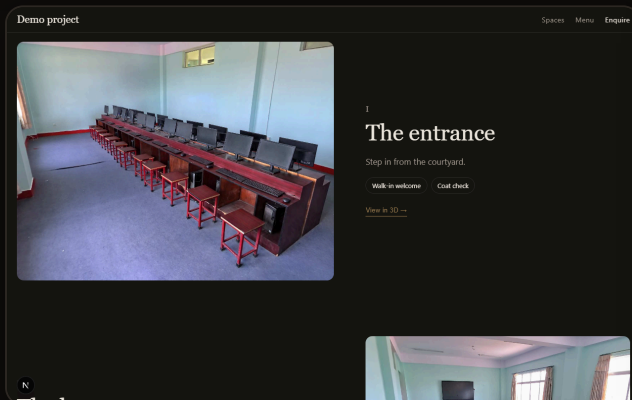
Each client gets a website at `/s/<project>`, edited in the studio. The splat tour stays exactly the same; the website adds the story around it.



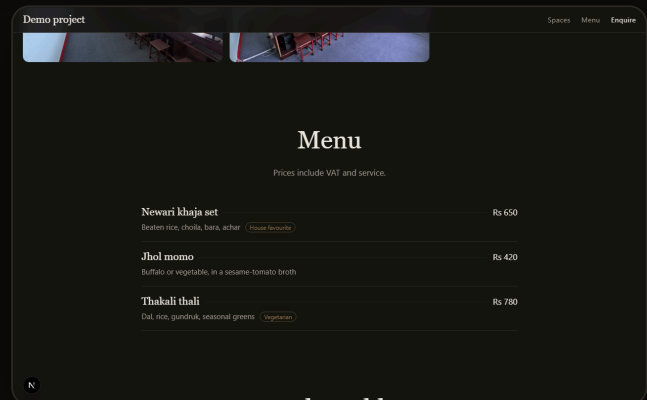
Hero in the client's font and colour.



The live tour, right on the page.



Rooms, each with "View in 3D".



Menu with prices and tags.

SECTIONS

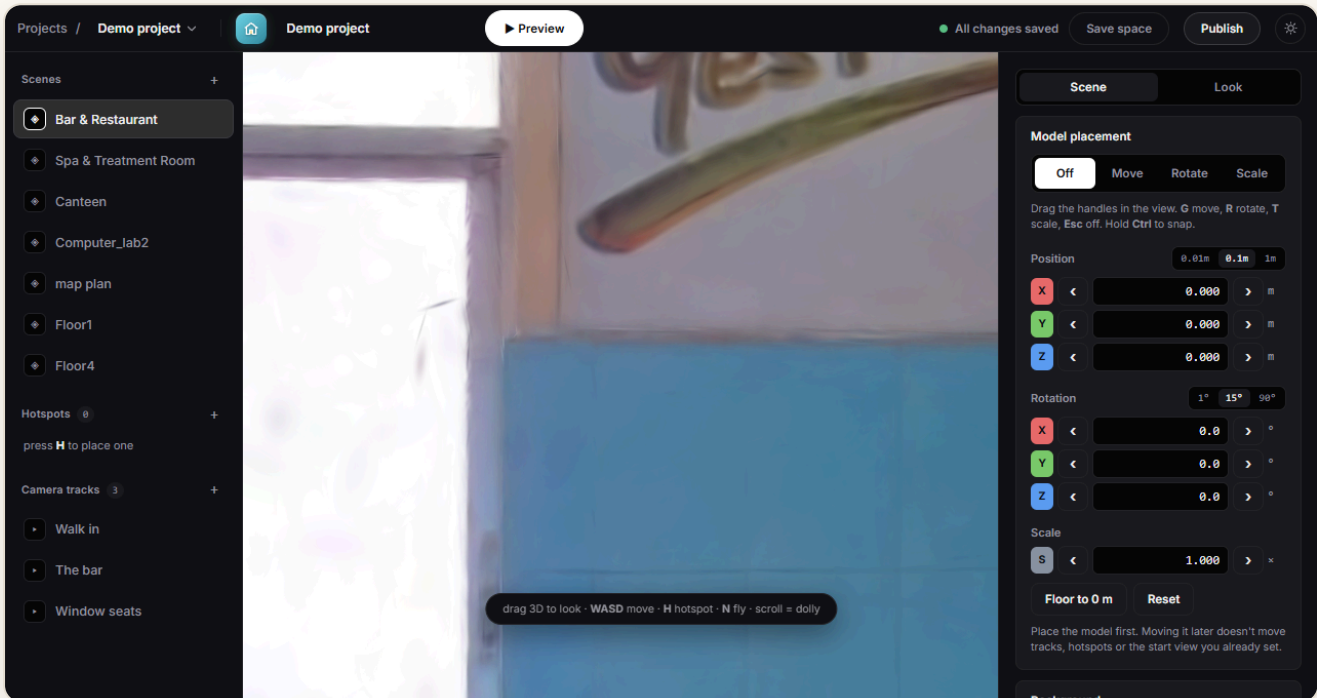
- ◆ **Live tour embed:** "View in 3D" flies it to each room
- ◆ **Facts, story, rooms** with features, floor plan and gallery
- ◆ **Menu** with prices, descriptions and tags
- ◆ **Book a table** and **Enquire**, plus the brand's phone, email, address, website and social links

EDITING AND PUBLISHING

- ◆ **Draft, Preview, Publish:** visitors only see what's published
- ◆ **Brand-aware:** logo, colour and heading font; empty sections fill from brand info
- ◆ **Readable text on any colour:** button text flips to black or white by contrast
- ◆ **Server-rendered** for search engines, with scroll reveals and dark mode

Where the team *builds* it all.

One project per client. Each project keeps its spaces, website, enquiries, reservations and reports together.



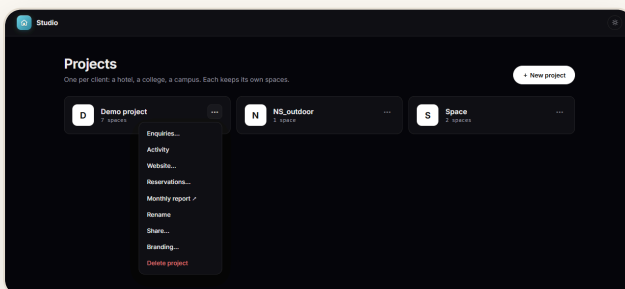
The scene editor. Scenes, hotspots and camera tracks on the left; model placement, look and Book now on the right.

SCENE EDITOR

- ◆ **Place the model:** move, rotate and scale with handles or exact numbers; snap; floor to 0 m
- ◆ **Hotspots** placed with one key (H), and viewpoints and camera tracks recorded in place
- ◆ **Autosave**, with a Preview of exactly what visitors will see

PUBLISHING YOU CAN UNDO

- ◆ **Publish, unpublish, revert** to the live version
- ◆ **Version history:** restore any earlier save
- ◆ **Activity log:** who changed what, and when
- ◆ **Night version** of a space and floor plans linked per space

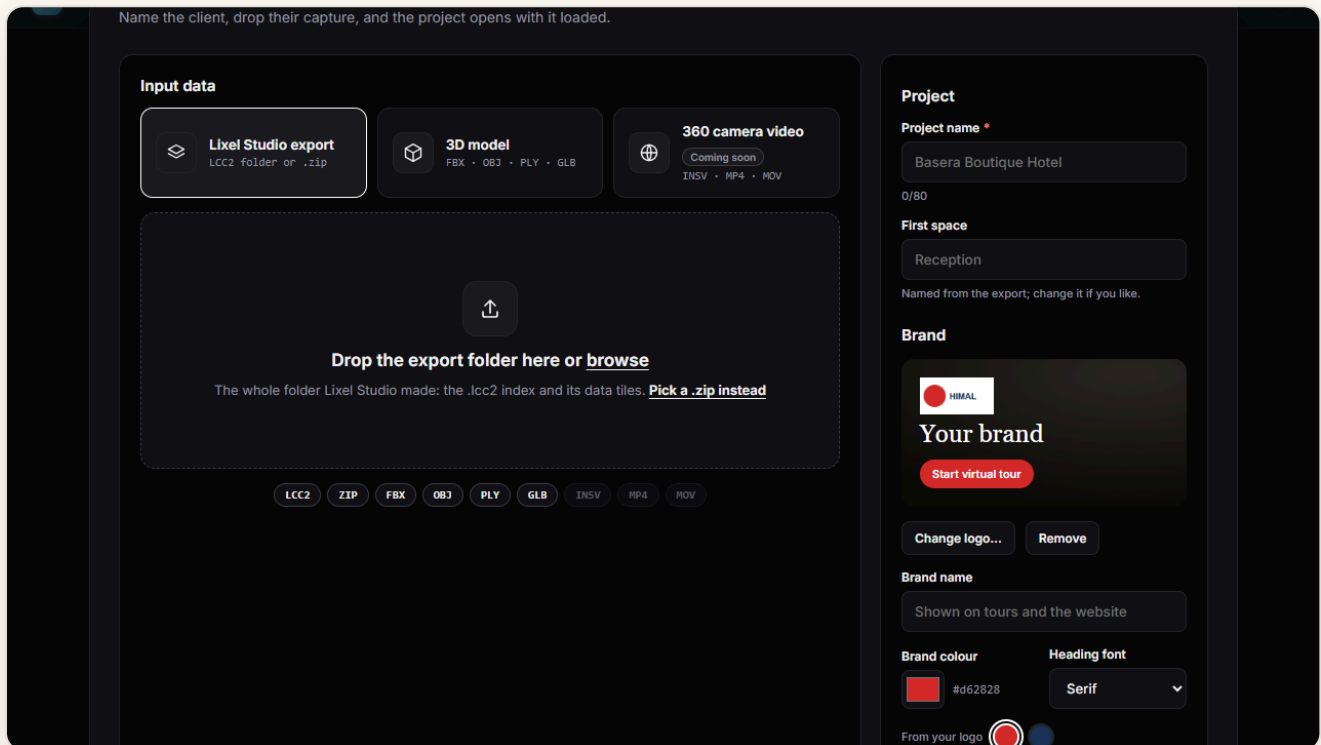


ONE MENU PER PROJECT

Enquiries, Activity, Website, Reservations, Monthly report, Rename, Share, Branding and Delete, all from the project card.

The brand, *read* from the logo.

Upload a client's logo when creating the project and the studio suggests the brand colour from the logo itself.



New project. A red and navy logo, and the red suggested as the brand colour, with every logo colour offered as a swatch.

HOW THE COLOUR IS CHOSEN

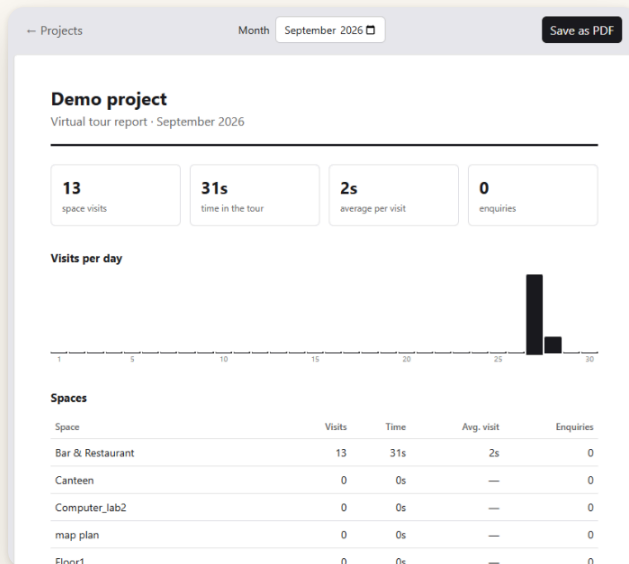
- ◇ The logo is read pixel by pixel, ignoring the background and anti-aliased edges
- ◇ Similar shades are grouped; vivid colours that cover more of the logo rank first
- ◇ A black-and-white logo suggests no colour at all; the default stays
- ◇ A colour picked by hand is never overwritten by a later logo

BRAND KIT PER PROJECT

- ◇ **Logo, brand name, colour, heading font** (serif, sans, classic)
- ◇ **Brand information:** tagline, about, phone, email, address, website, Instagram, Facebook
- ◇ Used by the tour, the hub page and the client website
- ◇ Editable any time from **Branding** in the project menu

Proof, on *paper*.

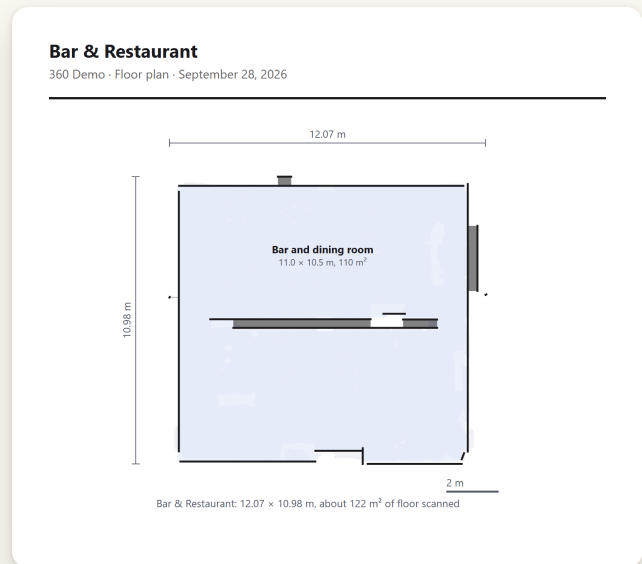
Two printable documents a client can keep: a monthly report of how the tour performed, and a measured floor plan drawn from the scan.



Monthly report. Visits, time in the tour, enquiries, visits per day and per space.

MONTHLY REPORT

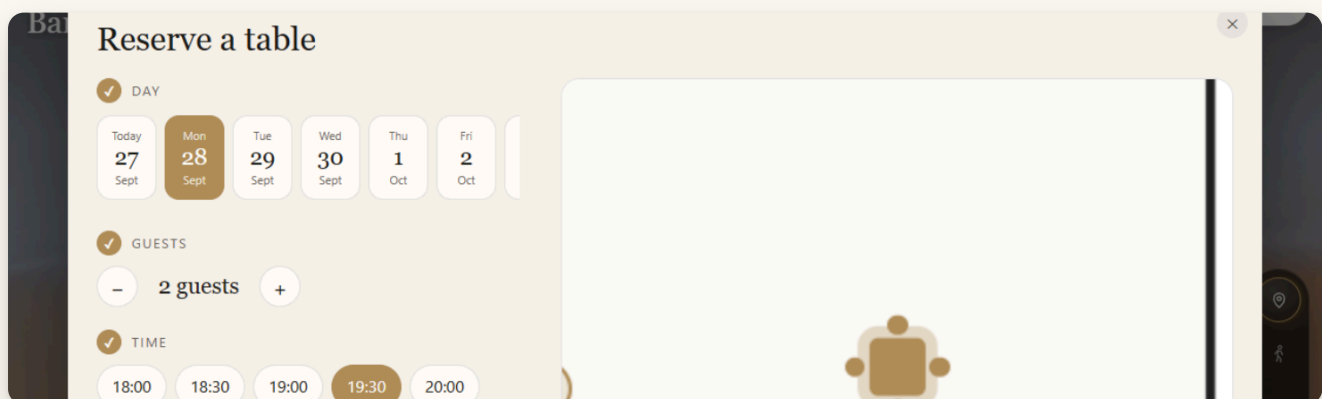
- ◇ Pick a month; Save as PDF from the browser
- ◇ Per-space table of visits, time, average visit and enquiries
- ◇ Enquiries also export as CSV



Floor plan sheet in ink on white, with named rooms and a schedule of sizes and areas.

FLOOR PLANS

- ◇ **Drawn from the scan** as a 2D or 3D-style plan, or **uploaded** by the client
- ◇ **Rooms, doors and windows** found on their own; rooms **named** in the studio keep their names
- ◇ Shown in the tour with "you are here" and on the website
- ◇ Prints **ink on white**, with a room schedule



Plans inside the tour. The restaurant's plan opens over the 3D view; "See the view from this table in 3D" flies the camera to that seat.

Your work is *yours*.

Two people on the same studio see only their own projects, plus the ones they've been invited to.

PRIVATE PROJECTS

A project is visible to its owner and members only. Unfiled spaces belong to whoever created them. Drafts need a signed-in owner or member.

SHARING

Invite teammates to a project by email. Older projects created before accounts can be claimed by their owner.

TEAM ROLES

Admin and editor roles, password reset links and removing accounts. The last admin can't be removed by mistake.

Who can do what

	OWNER	MEMBER	OTHER ACCOUNTS	PUBLIC
See the project in the studio	✓	✓	—	—
Edit spaces and the website draft	✓	✓	—	—
Read enquiries	✓	✓	—	—
Branding, rename, share, delete	✓	—	—	—
Visit the published tour and website	✓	✓	✓	✓

How the API protects it

LAYER	WHAT IT DOES
Sessions	Signed JWT sessions in HTTP-only cookies. Roles are checked fresh on every request, so a removed user loses access at once.
Access checks	Every project, scene, website and reservation route checks ownership or membership on the server, not just in the UI.
Spam protection	Rate limits on sign-in, sign-up and public forms; hidden honeypot fields and a minimum fill time on enquiries and table bookings.
Embeds	Embed keys work only on the websites the client lists.
Uploads	File paths are validated so an upload can never reach outside its own folder.
Secrets	Keys and passwords live only in <code>server/.env</code> , which is never committed.

Every scan, on *every* PC.

Scans can live in a cloud bucket instead of on the PC that uploaded them, so any computer, and the hosted site, shows the same splats.

A

Upload once

The studio uploads a scan; the API streams it straight into the bucket.

B

Served by the API

Visitors' browsers fetch through the API, which reads from the bucket. No bucket settings needed.

C

Seen everywhere

Any PC running the platform, or the hosted API, shows the same scans.

WHAT'S BUILT

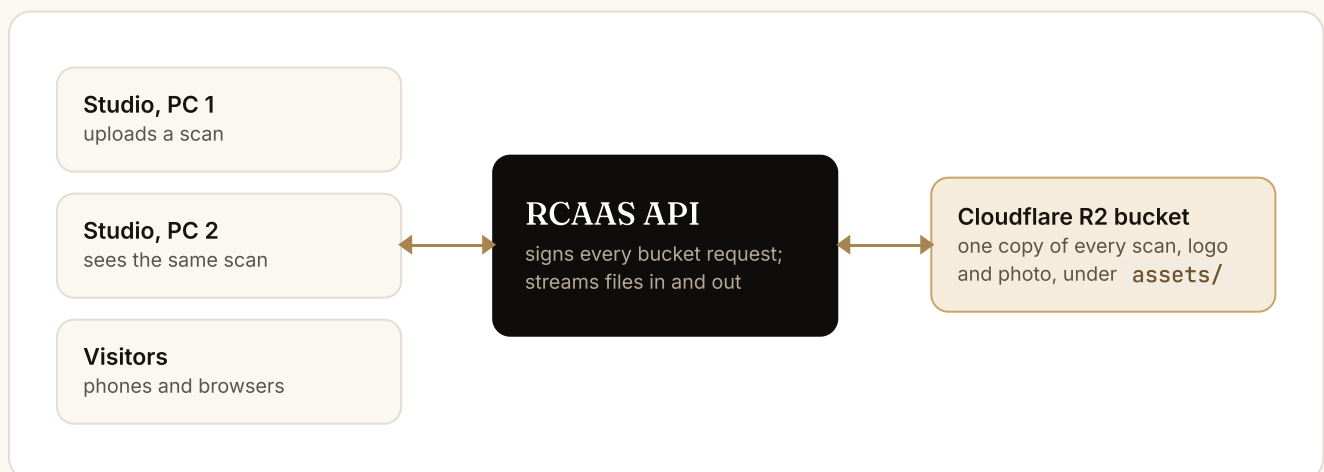
- ◆ **One switch:** `ASSET_DRIVER=local` or `s3` in `server/.env`
- ◆ **Any S3-compatible bucket:** Cloudflare R2 recommended (no download fees), also AWS S3, Backblaze B2 and MinIO
- ◆ **No extra libraries:** AWS request signing is written in-house and checked against AWS's published example
- ◆ **Large files stream;** partial (range) reads for fast scan loading
- ◆ **Floor plans** are drawn from scans stored in the bucket too

MOVING EXISTING SCANS

```
# on each PC that has uploads
cd server
npm run assets:push -- --dry-run
npm run assets:push
```

Copies what's missing, skips files already there, deletes nothing. Safe to run again.

The whole API test suite also runs against a simulated bucket, so both storage modes are tested on every run.



With `ASSET_DRIVER=s3`, nothing is stored on the PC that uploaded it. Bucket keys never leave the API.

Built to *last*.

PART TECHNOLOGY

Frontend	Next.js 16 (App Router, server components), React 19, TypeScript
3D	Three.js and React Three Fiber, Gaussian splats (LCC), 3D models and 360° video, BVH collisions for Walk mode
Motion	GSAP 3 (SplitText, ScrollTrigger) and Framer Motion
API	Express 4 with file-backed JSON stores; 84 endpoints
Email	Resend, for enquiry and reservation emails
Storage	Local disk, or any S3-compatible bucket (Cloudflare R2)

DEVELOPER EXPERIENCE

- ◇ **One command:** `npm run dev` starts the site and the API together
- ◇ **Self-healing ports:** leftover servers from a closed VS Code are stopped, and free ports are picked automatically
- ◇ **API recovers** if its port is taken

QUALITY CHECKS

- ◇ **130 automated tests:** 20 frontend, 110 API (one runs only with a real bucket)
- ◇ **Full API suite twice:** local disk and simulated bucket
- ◇ **TypeScript clean, lint clean** (no errors); production build of 22 routes
- ◇ **Checked in a real browser** at desktop and phone sizes

Getting started

```
npm install && (cd server && npm install)
cp server/.env.example server/.env    # then fill in secrets
npm run dev                            # site + API, ports picked for you
npm test                               # frontend + API tests
```

Where things live

<code>src/app/(site)</code>	RCAAS.tech marketing pages	<code>server/src/routes</code>	The API, one file per area
<code>src/app/t, s, tour</code>	Tour, hub page, client website	<code>server/src/storage*.js</code>	Local disk and S3/R2 storage
<code>src/app/studio</code>	Studio, website editor, reports	<code>server/test</code>	API tests, incl. a simulated bucket
<code>src/components</code>	Viewer, editor, booking, dialogs	<code>server/scripts</code>	<code>assets:push</code> for the bucket, <code>make-admin</code> for roles
<code>src/lib</code>	API client, i18n, brand colour, 3D helpers	<code>scripts/dev.mjs</code>	The <code>npm run dev</code> launcher

The road *ahead*.

What's built is ready to demo and to use. These are the steps to take it to production for every client.

STEP	WHY IT MATTERS
Host the API once	Railway or Render with the R2 bucket, so projects, websites and bookings are shared, not only the scans.
Turn on the bucket	Create the R2 bucket, set <code>ASSET_DRIVER=s3</code> , and run <code>assets:push</code> on each PC with uploads.
Set up email	Add <code>RESEND_API_KEY</code> so enquiry and reservation emails are sent.
Claim projects, name an admin	Claim the existing projects in the studio, then name the first admin with <code>npm run make-admin -- <email> in server/</code> .
Nepali review	A native speaker should review the नेपाली labels before launch.
INSV straight from the camera	Raw Insta360 files are exported in Insta360 Studio first; stitching them on upload would save that step.
Sound in 360° video	360° spaces play muted; a tap-to-unmute button would bring the sound in.
Windows behind curtains	Plans find doors and windows from the scan, but miss windows covered by curtains or blinds.

NEW IN THIS EDITION

360° video spaces, rooms with doors and windows drawn on floor plans, room names set in the studio, plans that print ink on white, and a `make-admin` script for naming the first admin.

*We walk your space once.
Your visitors walk it for years.*

RCAAS.tech · GeoNova Solutions, Kathmandu

Since the first *edition*

Everything added since this book was first printed: booking from inside the 3D space, events, a fuller client website, a calmer opening, and the groundwork to go live.

12	Booking from the 3D space	20
	Tables, rooms and halls, booked from their hotspot	
13	Weddings and events	21
	Halls by the part of the day, with deposits	
14	The client website, grown	22
	Three looks, offers, reviews, FAQ and a map	
15	More places, more kinds of place	23
	Several dining places, and a starter for every venue	
16	The tour and studio, refined	24
	The opening, collision boxes, exact values	
17	Contact and About, redesigned	25
	A field sheet that writes the email for you	
18	Accounts and data	26
	A real database, password reset by email, safer bookings	
19	Found online	27
	Search, answer engines, AI assistants and the policies	

91

API endpoints

148

automated tests

9

hotspot types

6

kinds of place to start from

3

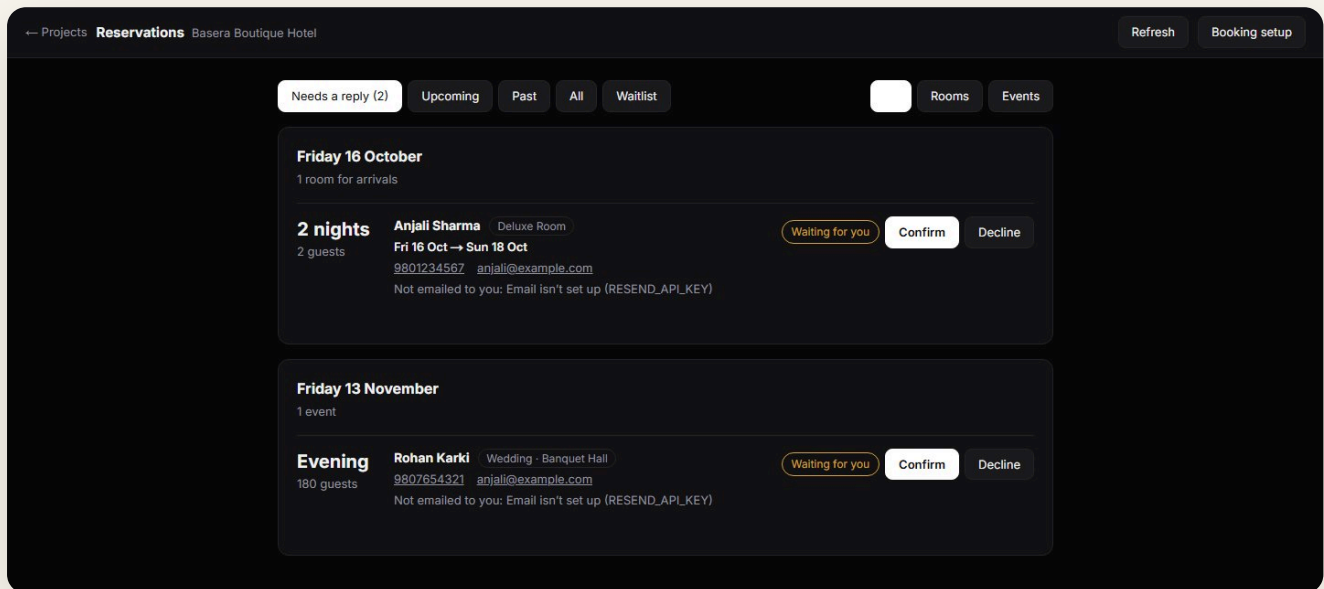
website looks

4

policy pages

Book the table *you can see.*

Tables, rooms and halls each carry their own Book now, so a guest books the very thing they are looking at in the scan, with nothing else set up first.



Reservations. A room request and a wedding request, straight from the guest, waiting for Confirm or Decline.

TABLES

Every table in a dining room is its own hotspot, numbered for you, with "Next table" placing the next one where you look. Booked by the sitting, with taken times struck out.

ROOMS

Dates and guests on the room itself, with the price and deposit you typed. Or straight to the hotel's own booking page, dates filled in.

HALLS

Daytime, evening or the whole day. A part of the day already asked for is struck out, and a clash is refused, so a hall is never promised twice.

The hall, *by the evening.*

Banquet halls are booked the way venues sell them: by the part of the day, for a kind of event, with the deposit stated up front.

CELEBRATE WITH US

Weddings and events

Pick a hall, the day and how many guests. We'll come back to you about everything else.

1 HALL

Pick a hall from the list.

2 DAY

The day of your event

mm/dd/yyyy

Daytime Evening Whole day

3 GUESTS AND EVENT

How many guests What kind of event

Choose one

We hold a date for three days while you decide.



Banquet Hall

Seats 150 - 250 standing · 320 m²

From Rs 1,500 per plate

Deposit: Rs 50,000 to hold the date

Stage dance floor own entrance

See it in 3D →

Weddings and events, on the client's website: pick a hall, a day and daytime, evening or the whole day.

EVERY HALL

Seated and standing capacity, size, a photo, its own 3D space, and a price and deposit in plain words.

NO DOUBLE BOOKING

The whole day clashes with either half; the halves don't clash with each other. Requests wait in Reservations.

IN THE TOUR TOO

"Plan an event" in any space, and "Book this hall" inside the hall itself. Every request is counted in the monthly report.

One website, *three moods.*

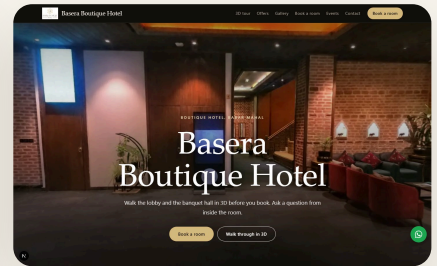
Each client's website now has a look to match the place, and the sections that turn a visitor into a guest.



Heritage



Modern



Night

The same website in its three looks, each in the client's own colour and font. The look no longer follows the visitor's dark mode.

OFFERS AND PACKAGES

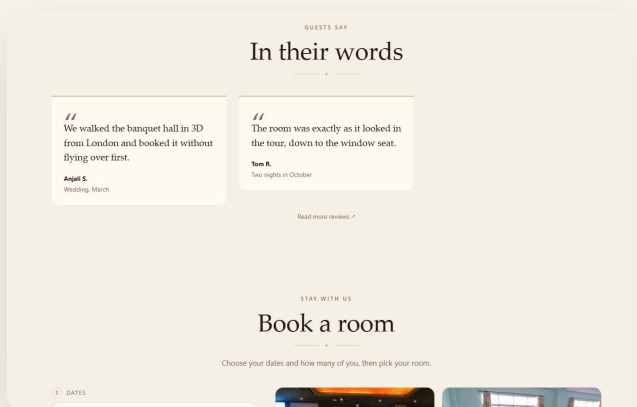
Each with "Ask about this offer", which opens an enquiry already naming it.

GUEST REVIEWS AND FAQ

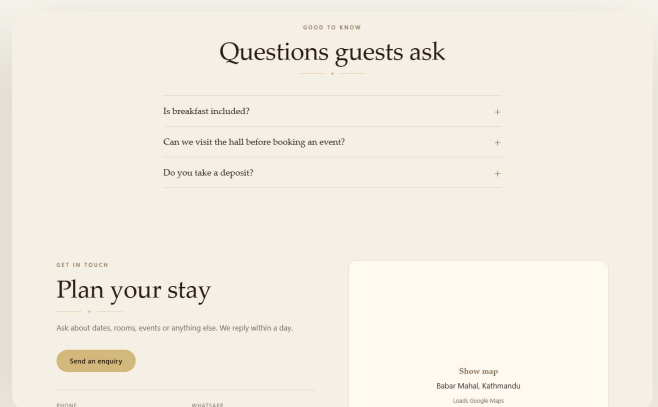
Quotes with a link to read more, and questions answered on the page and for search engines.

PHOTOS, MAP, WHATSAPP

Full-width photos up to 25 MB served small and sharp, a gallery that opens large, a Google map on request, and a WhatsApp button.



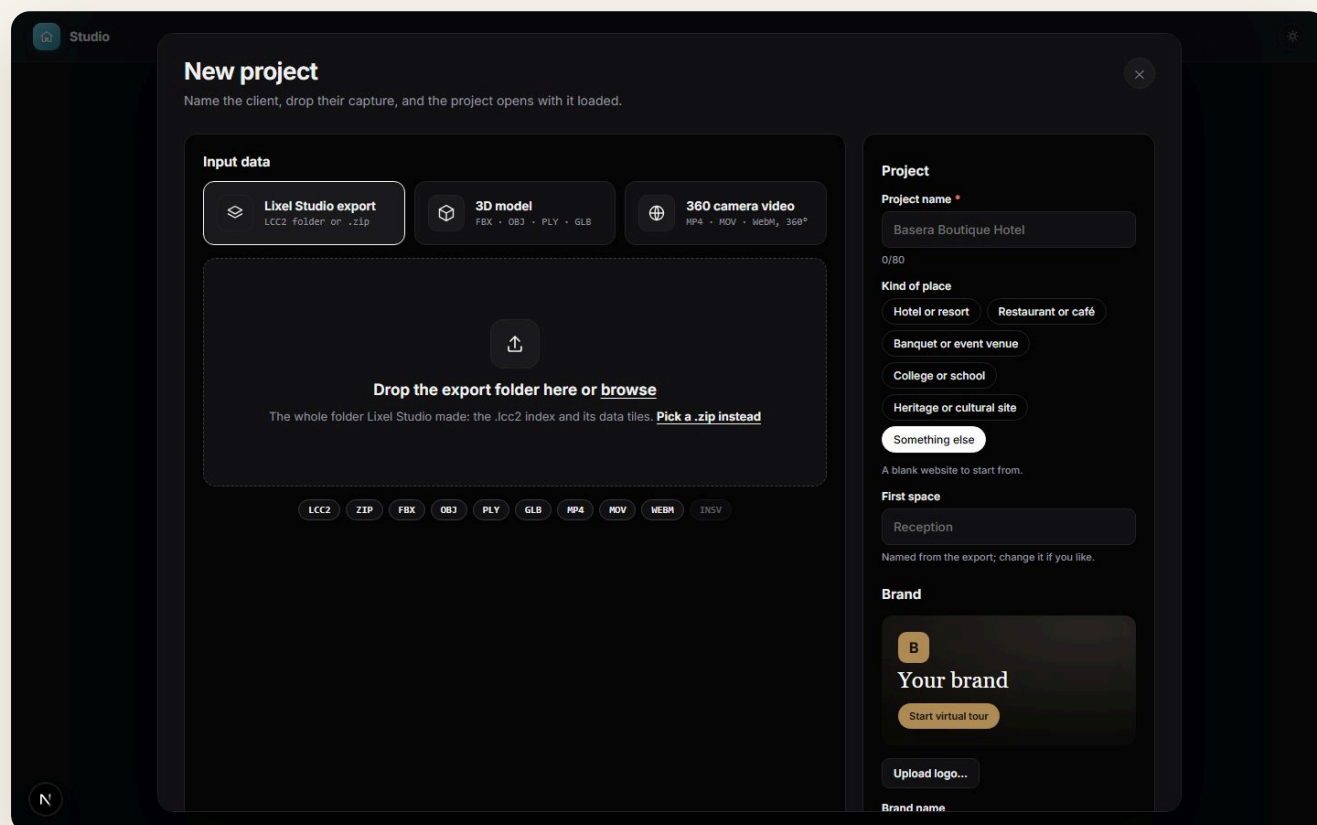
Guest reviews, in the guests' own words.



Questions, answered before anyone has to ask.

A café, a bar, *a banquet venue*.

A hotel rarely has one place to eat, and not every client is a hotel. Both are now built in.



New project. Pick the kind of place, and the website starts with the right sections, switched off until they are filled in.

KINDS OF PLACE

Hotel or resort, restaurant or café, banquet venue, college, heritage site, or something else. A hotel starts with rooms, tables, a menu and a hall.

MORE DINING PLACES

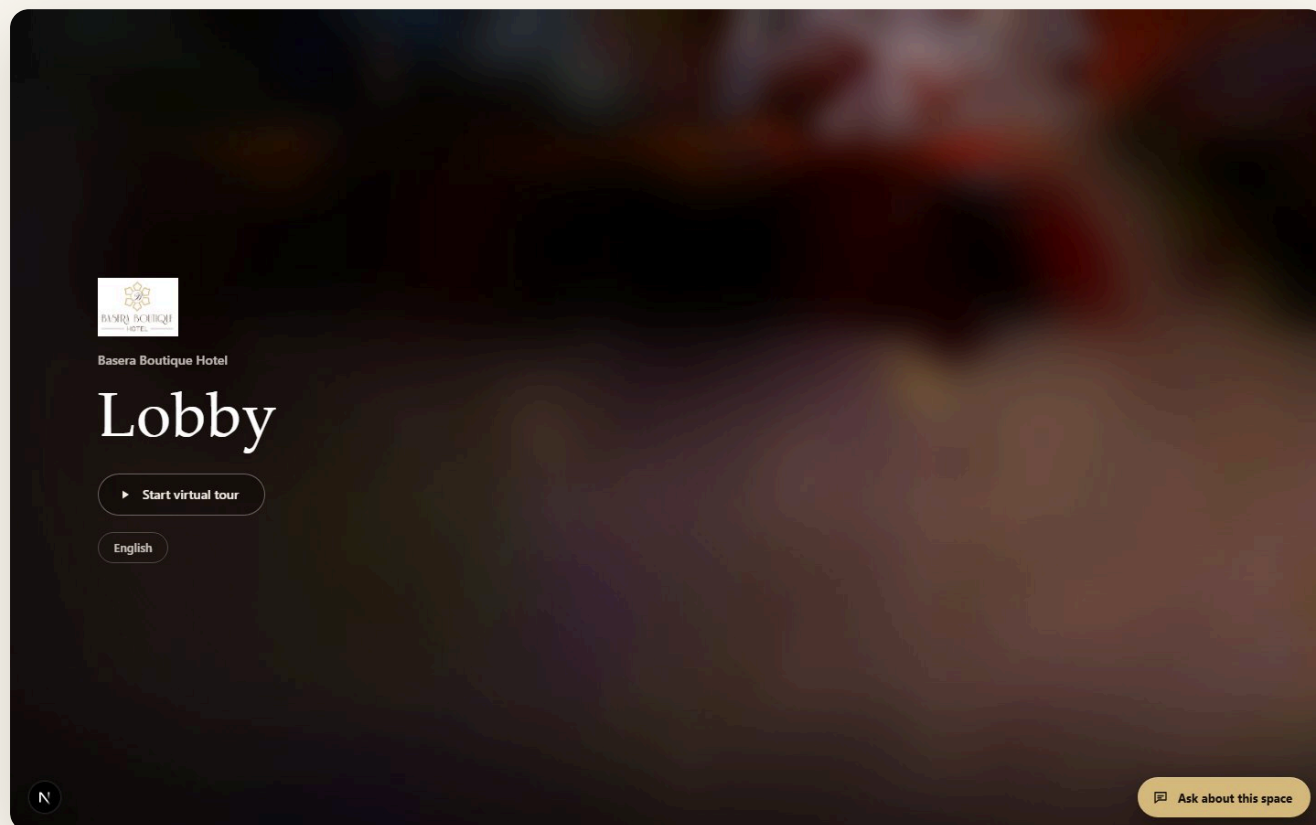
A café beside the restaurant, a rooftop bar: each with its own name, menu, floor plan, tables and hours, and its own "T1".

BOOKED APART

In the tour, a table books only at its own place. With more than one place taking bookings, guests pick the place from a list.

A calmer *first step*.

The room now streams in behind frosted glass while the visitor reads the welcome, then the camera glides to the start view.



The opening. The space loads behind the start screen; on Start the glass dissolves and the camera glides in. Phones get a lighter veil, and reduced motion gets no glide.

COLLISION BOXES

Invisible walls and floors for holes in a scan, placed, sized and turned in the studio, saved with undo, and published with the space.

EXACT VALUES

Every studio slider takes a typed value: a camera track's length up to 60 seconds, eye level and clearance in metres.

SMALL THINGS

Hotspots land on the surface you aim at. Each waypoint has Go and Set here. Embeds go full screen on Start. A quality drop keeps the visitor's place.

Pages from a *field book*.

The company's own pages now read like a surveyor's field book: ruled answer lines, a register of measured sites, and contour lines that draw themselves in.

Contact. Five answers on a ruled sheet. "Write the email" opens the visitor's own email app with them filled in, beside the phone number and the office on a contour map.

CONTACT

What a crew needs before a site visit: the space, where, how big, what's needed, and a phone to call back.

ABOUT

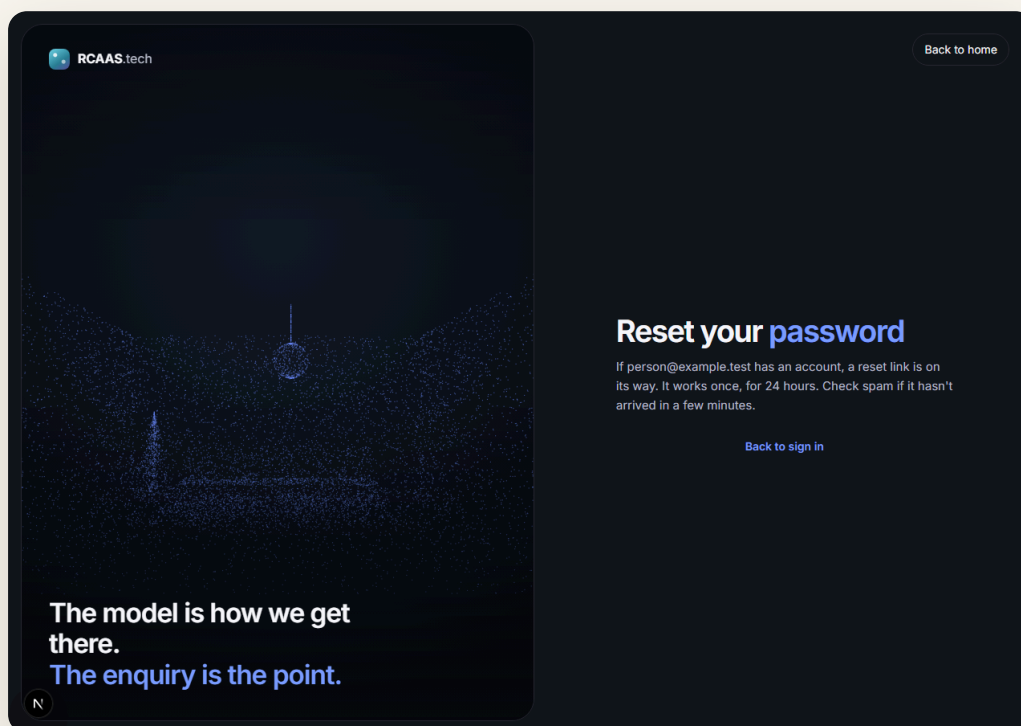
A register of the sites measured, from the Gwarko Overpass to Basera, the mission, the kit and the partners.

PAGE NOT FOUND

A missing link now lands on a page with the way back home, instead of a bare "404".

Ready for a *real database*.

Studio accounts can now live in Postgres (Supabase), the first step off files. Moving to another provider later is one export and one address.



Forgot password. A one-time link by email, valid for a day. The answer is the same whether or not the email has an account.

SAFER BOOKINGS

Halls and tables refuse a clash, and taking back a declined booking checks again first, so nothing is booked twice.

MISSING SCANS, CAUGHT

The studio warns before publishing a space whose scan files aren't on the server, instead of a tour that never shows its 3D.

FOR DEVELOPERS

A Developer Guide beside this book: the architecture, all 91 API routes and every file, rebuilt from the code with one command.

Easy to find, *easy to trust.*

The site now tells search engines, answer boxes and AI assistants exactly what RCAAS.tech is, and tells visitors exactly what happens to their data.

SEARCH

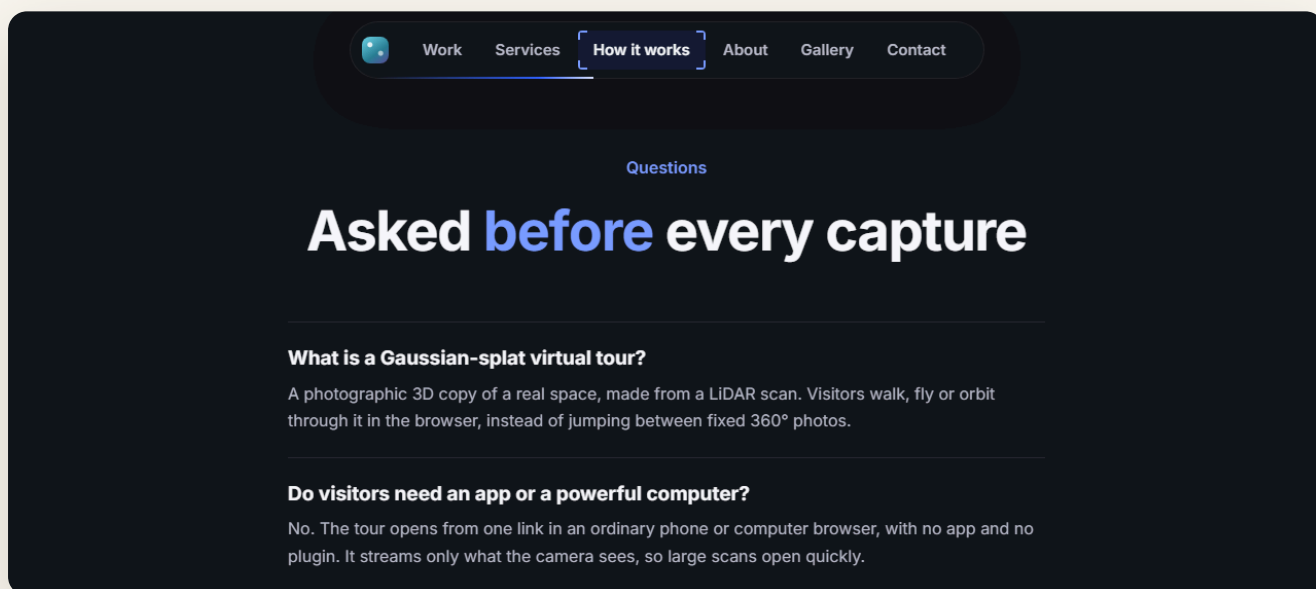
A sitemap that lists every published website and tour, canonical links on every page, and private pages kept out of results.

ANSWER ENGINES

A visible FAQ on How it works, marked up so search can quote the answers directly.

AI ASSISTANTS

The business and its services as structured data on every page, and an llms.txt summary for ChatGPT, Claude and Perplexity.



How it works. Nine questions buyers ask, answered on the page and marked up for search and AI answers.

TERMS OF SERVICE

What visitors and clients agree to, written plainly.

PRIVACY AND SECURITY

What is collected, who sees it, how accounts are protected, and how to report a problem.

COOKIE POLICY

One cookie, only when signing in. No tracking, and the client's map loads only on request.

Drafts written to match what the code does. They need a lawyer's review before launch.